Wildfire Survival Prediction

AI-Powered Community Risk Intelligence · WiDS Global Datathon 2026

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Houston City College | ITAI 2377 Data Science in Artificial Intelligence | Spring 2026

THE CHALLENGE

Will This Wildfire Reach Your Neighborhood?

When a wildfire ignites, emergency coordinators face one urgent question: **how much time do we have?** We built an AI system that answers it — with calibrated, probability-based timelines across four critical decision windows.

12h	24h	48h	72h
IMMEDIATE	NEAR-TERM	PLANNING	STRATEGIC

> *The output is not a yes/no alarm — it's a probability curve. Responders get **"72% chance of impact within 24 hours"** instead of just "fire nearby." That difference is what enables proportional, intelligent response.*

Outputs: **prob_12h · prob_24h · prob_48h · prob_72h** per fire event — submitted to the WiDS 2026 Kaggle competition

THE DATA REALITY

221 Events. Real Consequences.

221
HISTORICAL FIRE TRAJECTORIES

~31%
REACHED THE COMMUNITY ZONE

83
RAW METEOROLOGICAL FEATURES

Analogy: *Imagine predicting airplane crash risk using only 221 total flight records — 69 of which crashed. Standard AI would memorize those 69 crashes. On a new flight it's never seen, it fails.*

The hidden challenge: Not every fire's story ends on record — fires stopped by weather or containment crews are *right-censored.* Standard classifiers discard this. Survival analysis extracts signal from it. Those "near misses" tell us exactly what slowed the fire down.

THE CRITICAL DECISION

Why We Didn't Use Standard Deep Learning

✗ DEEP LEARNING / CLASSIFICATION

With only 221 records, a neural network **memorizes the training data.** It learns specific fire coordinates, not universal fire physics. On a new wildfire in a new location — it collapses.

"Like memorizing the answer key instead of learning the subject."

✓ KINEMATIC SURVIVAL ANALYSIS

Instead of memorizing events, we **model the physics of fire spread.** Distance. Speed. Direction. These laws apply everywhere, regardless of geography. The model generalizes.

"Like learning the rules of the road instead of memorizing one specific map."

Key insight: Survival analysis was built for exactly this scenario — small N, censored data, time-to-event prediction. It was the right tool for the job from the start.

FEATURE ENGINEERING

We Taught the Model to See Fire Like a Physicist

Physics-Informed Feature Engineering

```
def load_and_engineer_features(train, test):
    ...
    # Feature Engineering: Physics Interactions
    for df in [X_train, X_test]:
        # Danger Index: How fast it's closing * how aligned it is
        df['danger_index'] = df['closing_speed_m_per_h'] * df['alignment_cos']

        # Time to impact estimate
        safe_close_speed = df['closing_speed_m_per_h'].clip(lower=0.1)
        df['time_to_impact_est'] = df['dist_min_ci_0_5h'] / safe_close_speed

        # Growth impact metrics
        df['growth_impact'] = df['area_growth_rate_ha_per_h'] * df['alignment_cos']
        df['dist_accel_impact'] = df['dist_accel_m_per_h2'] / safe_close_speed

    return X_train, y_train_struct, X_test
```

Raw sensor data gives us measurements. Physics gives us *meaning.*

DANGER INDEX

How aggressively is the fire closing in?

`closing_speed × alignment_cos`

TRAJECTORY ALIGNMENT

Is the fire pointed directly at the community?

`cos(θ) → +1 = direct hit course`

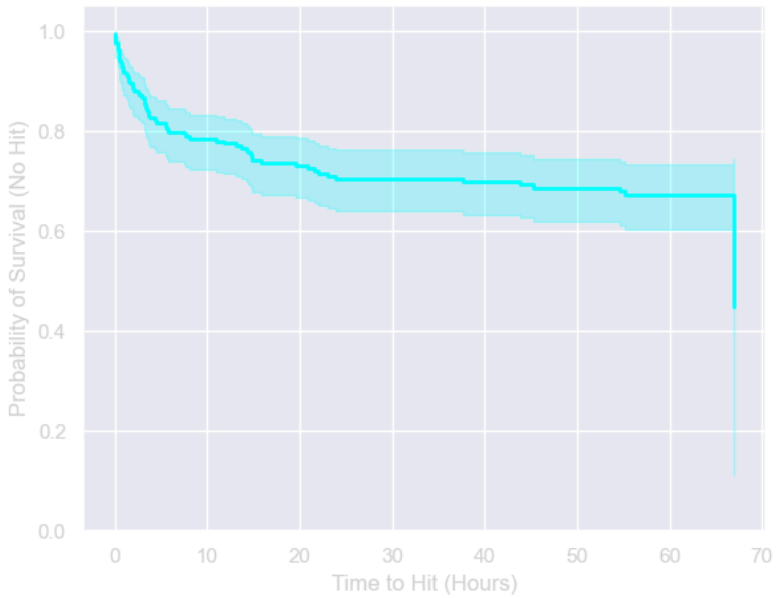
TIME-TO-IMPACT

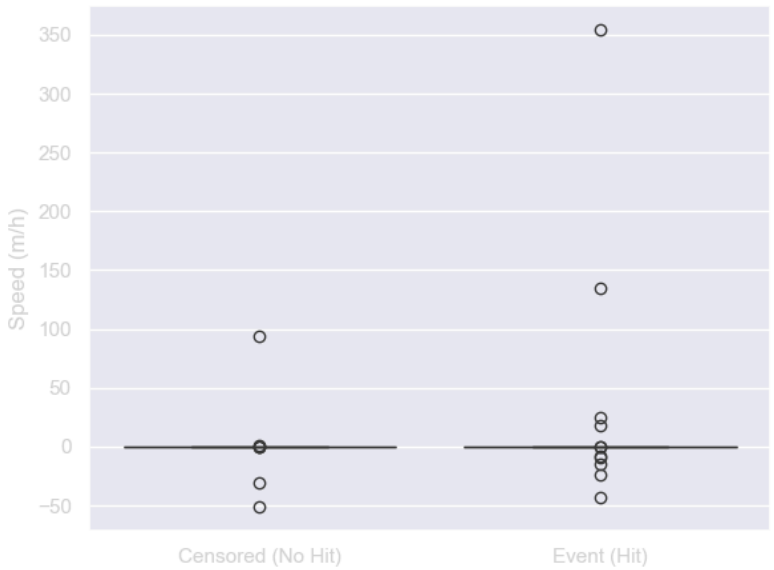
Estimated hours until the fire arrives

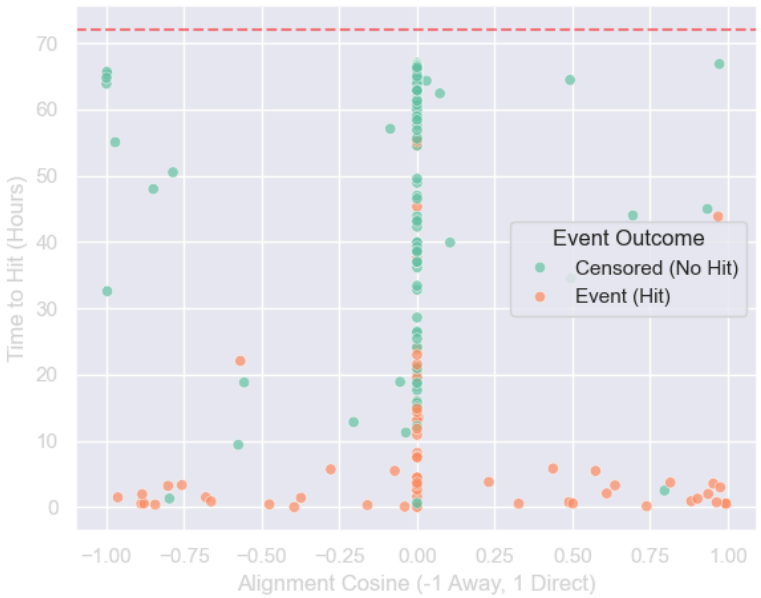
`distance ÷ closing_speed`

EXPLORATORY DATA ANALYSIS

The Data Confirmed Our Physics Hypothesis







Survival curve: Most fires resolve within 30 hours, validating our four time-horizon selection.

Trajectory alignment: Hit events cluster near cosine ≥ 0 — fires pointed at communities reach them.

Closing speed: Hit events show clearly higher median speed, confirming the Danger Index has real signal.

MODEL ARCHITECTURE

Two Experts. One Verdict.

Neither model alone was enough. Together, they achieve what neither could do solo.

RANDOM SURVIVAL FOREST

The Seasoned Ranger

Builds hundreds of survival decision trees. Naturally resistant to noise and small-sample overfitting. Captures complex, non-linear fire kinematic interactions.

Native cumulative hazard splitting · High-variance resistance

+

50/50 blend

GRADIENT BOOSTING SURVIVAL

The Precision Analyst

Systematically corrects every residual error from the prior round. Bayesian-tuned across 50 Optuna trials for optimal depth, learning rate, and leaf constraints.

50-trial Optuna TPE · max_depth=3 · min_samples_leaf=14

Final output: equally weighted average of RSF and GBSA survival probability distributions

VALIDATION INTEGRITY

We Made Sure the Score Was Real

STRATIFIED K-FOLD (5 SPLITS)

With only ~31% hit-events, a random split could accidentally exclude *all* disaster cases from a test fold — making the score meaningless. We locked every fold to mirror the exact 31% hit ratio.

"Every test set looks like the real world."

MONOTONIC PROBABILITY BOUNDS

A fire cannot have a *higher* 12-hour probability than 72-hour. We enforce strict ordering with a custom epsilon buffer to prevent Brier Score metric collapse at survival boundaries.

"P(12h) ≤ P(24h) ≤ P(48h) ≤ P(72h). Always."

Zero-Leakage Stratified Validation Strategy

```
# ================================================================
# [STEP 4] Stratified Cross Validation Strategy
# ================================================================
N_SPLITS = 5

# Using Stratified K-Fold to prevent catastrophic label imbalance
# where extremely dense survival events bleed randomly into test blocks
skf = StratifiedKFold(n_splits=N_SPLITS, shuffle=True, random_state=RANDOM_SEED)

stratify_key = y_train_df['event'].astype(int)
cv_splits = list(skf.split(X_train, stratify_key))

print("Total Splits verified against data leakage.")
```

RESULTS

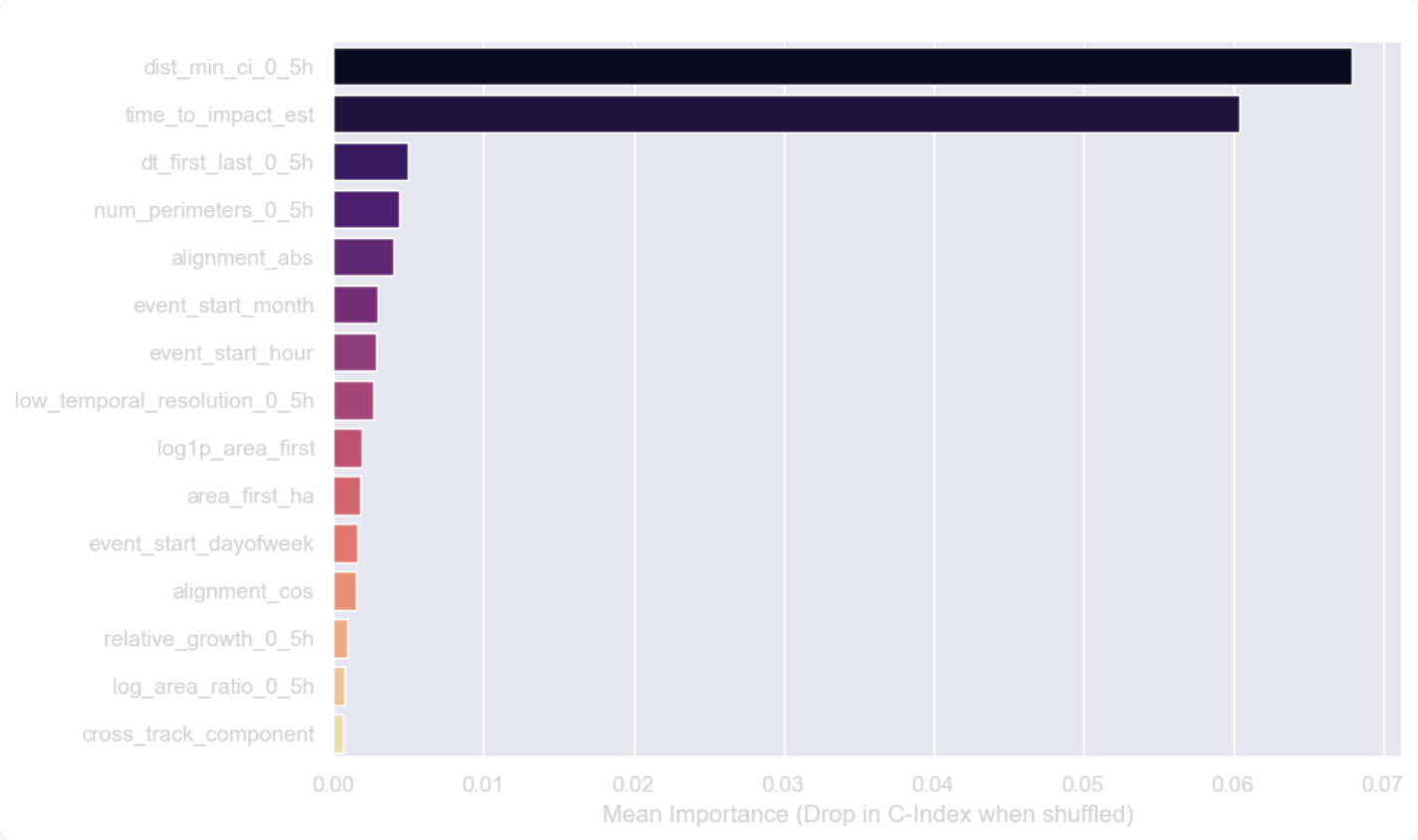
0.9737

Hybrid Concordance / Brier Score

- Top-tier out-of-fold CV performance
- Engineered features beat raw sensors
- Zero data leakage confirmed
- Probabilistically valid outputs

Concordance → ranking accuracy.
Brier Score → probability calibration.
Both matter for emergency response.

Feature Importance (Permutation)



dist_min_ci_0_5h and **time_to_impact_est** — both features we engineered — dominate all 83 raw inputs. Physics wins.

ACTIONABLE INTELLIGENCE

Precise Probability Windows for Every Fire

Each fire event receives four calibrated risk probabilities — not a binary alert, but a **decision timeline** emergency coordinators can act on proportionally.

Final Survival Probabilities (First 10 Events)

	event_id	prob_12h	prob_24h	prob_48h	prob_72h
0	10662602.0	0.014884639774200	0.031213960728970	0.036731365138990	0.080674941581567
1	13353600.0	0.597326791435590	0.903784621044740	0.926461954202290	0.950626776425088
2	13942327.0	0.014888690125459	0.028552229264720	0.035594956906440	0.071032627863219
3	16112781.0	0.634911451032550	0.900881115812050	0.915750434700460	0.939638458227518
4	17132808.0	0.328810218987120	0.634907554988608	0.359283499945479	0.407444007632913
5	17445696.0	0.008442741840260	0.016793533184550	0.020637580699560	0.050159778023788
6	17599982.0	0.015064039816520	0.031214213287190	0.036510361668540	0.073980559674157
7	18750374.0	0.282007545952810	0.631702687533290	0.754847516522230	0.890870150868375
8	21365245.0	0.008408176478900	0.015973755001720	0.020162895701540	0.048395844382385
9	23634840.0	0.478526278235020	0.850286104104710	0.891177093163380	0.942742641907501

WHAT THIS PROJECT DEMONSTRATES

Technical Depth. Real-World Judgment.

This was a live global competition with real data constraints — not a classroom exercise.

TECHNICAL SKILLS

- Survival analysis (RSF + GBSA)
- Physics-informed feature engineering
- Bayesian hyperparameter optimization
- Zero-leakage cross-validation design
- Censored data handling
- End-to-end ML pipeline (ETL → model → output)

JUDGMENT & THINKING

- Chose the right tool for a constrained problem
- Translated domain physics into ML features
- Prioritized trust and interpretability over raw score
- Built for emergency responders, not just Kaggle
- Competed in a global academic datathon

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scikit-survival | XGBoost / LightGBM | Optuna | pandas / numpy | Stratified K-Fold | Permutation Importance | Python | Jupyter

WIDS GLOBAL DATATHON 2026

Thank You

Questions?

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Hybrid Score: **0.9737** · Kinematic Survival Analysis